Low-field carrier mobility models for 3D electron gas (3DEG) slab

Badal Mondal

February 3, 2026

This documentation is part of `mobilitypy` package.[1] The following models for low-field electron mobility (μ) calculations from different scattering mechanism are based on the Reference-[2, 3, 4, 5, 6, 7] and the references therein.

The available mobility models are:

- 'v1' (available in mobilitypy-v1.0.5 onwards)

---

1. Total mobility:

$$\mu = \left(\sum_i \frac{1}{\mu_i} \right)^{-1}$$

   where μ_i's are the mobility from different scattering mechanisms presented below.

2. The mobility models here allows to calculate position dependent mobility, i.e. $\mu(z)$.

1 Parameters

Parameters name	Symbol	SI Unit
Electron elementary charge	e	C
Boltzmann constant	k_B	$\mathrm{J\,K^{-1}}$
Temperature	T	K
Mass density	ρ	$\mathrm{kg\,m^{-3}}$
Lattice parameter	a_0	Å
Lattice parameter	c_0	Å
Unit cell volume	Ω_0	$\mathrm{Å}^3$
Bandgap energy	E_g	eV
Alloy-disordered scattering potential	U_0	eV
Electron effective mass	m^*	m_0
Electron rest mass	m_0	kg
Static dielectric constant	ϵ_s	ϵ_0
High frequency dielectric constant	ϵ_h	ϵ_0
Vacuum permittivity	ϵ_0	$\mathrm{C\,V^{-1}\,m^{-1}}$
Longitudinal acoustic phonon velocity	v_{LA}	$\mathrm{m\,s^{-1}}$
Transversal acoustic phonon velocity	v_{TA}	$\mathrm{m\,s^{-1}}$
Conduction band deformation potential	E_D	eV
Polar optical phonon energy	E_{POP}	eV
Electromechanical coupling coefficient	K^2	-
Elastic constants	C_{ij}	GPa
Fermi wave vector	k_F	$\mathrm{cm^{-1}}$
Thomas-Fermi wave vector	q_{TF}	$\mathrm{cm^{-1}}$
POP wave vector	k_0	$\mathrm{cm^{-1}}$
Volumetric electron gas density	n_{3d}	$10^{18}\ \mathrm{cm^{-3}}$
Dislocation density	N_{DIS}	$10^{8}\ \mathrm{cm^{-2}}$
Dislocation occupancy	f_{DIS}	-
Mobility	μ	$\mathrm{cm^{-2}\,V^{-1}\,s^{-1}}$

Note: Proper unit conversions for all parameters have been implemented within the equations.

2 Basic theory of mobility

- **Fermi-Dirac function for fermions:**

$$f_0(E) = \frac{1}{1 + e^{\frac{E-E_F}{k_B T}}} \tag{1}$$

where E_F is the Fermi energy.

- **Boltzmann Transport Equation (BTE)**: Within relaxation time approximation:

$$f(\vec{k}) = f_0(\vec{k}) + e\mathbb{F}_i \tau(k) v_i \frac{\partial f_0}{\partial \varepsilon} \tag{2}$$

where $f(\vec{k})$ is carrier distribution function, $\tau(k)$ is the momentum relaxation time, v_i is the carriers velocity in the i-th direction in reponse of the field $\mathbb{F}_i$.

- **DOS in d-dimension:**

$$g_d(E) = \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} (E - E_C)^{\frac{d}{2}-1} \tag{3}$$

- **Total number of carriers/unit 'volume' in d-dimensional space:**

$$\begin{aligned}
n_{dd} &= \int \frac{d^d k}{(2\pi)^d} f(k) \\
&= \int_{E_C}^{\infty} f_0(E)\, g_d(E)\, dE \\
&= \int_{E_C}^{\infty} \frac{1}{1 + e^{(E-E_F)/k_B T}}\, g_d(E) dE \\
&= \int_{E_C}^{\infty} \frac{1}{1 + e^{(E-E_F)/k_B T}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} (E - E_C)^{\frac{d}{2}-1} dE \\
&= \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \int_{E_C}^{\infty} \frac{1}{1 + e^{(E-E_F)/k_B T}} (E - E_C)^{\frac{d}{2}-1} dE \\
&= \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \int_{E_C}^{\infty} \frac{1}{1 + e^{\frac{(E-E_C)-(E_F-E_C)}{k_B T}}} \left(\frac{E - E_C}{k_B T}\right)^{\frac{d}{2}-1} (k_B T)^{\frac{d}{2}-1}\, dE \\
&= \left(\frac{2m^*\, k_B T}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \int_{E_C}^{\infty} \frac{1}{1 + e^{\frac{(E-E_C)-(E_F-E_C)}{k_B T}}} \left(\frac{E - E_C}{k_B T}\right)^{\frac{d}{2}-1} \frac{1}{k_B T}\, dE
\end{aligned} \tag{4}$$

- **Fermi-Dirac integral of j-th order:**

$$F_j(\zeta) = \frac{1}{\Gamma(j+1)} \int_0^\infty \frac{u^j}{1+e^{u-\zeta}}\, du$$

- Substituting $\varepsilon = \frac{E-E_C}{k_B\,T}$ and $\eta_{_F} = \frac{E_F-E_C}{k_b\,T}$

$$f_0(\varepsilon) = \frac{1}{1+e^{\varepsilon-\eta_{_F}}} \tag{5}$$

$$F_{\frac{d}{2}-1}(\eta_{_F}) = \frac{1}{\Gamma(\frac{d}{2})} \int_0^\infty \frac{\varepsilon^{\frac{d}{2}-1}}{1+e^{\varepsilon-\eta_{_F}}}\, d\varepsilon = \frac{1}{\Gamma(\frac{d}{2})} \int_0^\infty f_0(\varepsilon)\, \varepsilon^{\frac{d}{2}-1}\, d\varepsilon \tag{6}$$

$$\begin{aligned} n_{dd} &= \left(\frac{2m^*\,k_BT}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}} \frac{1}{\Gamma(\frac{d}{2})} \int_0^\infty \frac{1}{1+e^{\varepsilon-\eta_{_F}}} \varepsilon^{\frac{d}{2}-1}\, d\varepsilon \\ &= \left(\frac{2m^*\,k_BT}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}}\, F_{\frac{d}{2}-1}(\eta_{_F}) \end{aligned} \tag{7}$$

- **Current:**

$$J_i = en \underbrace{\left(-\frac{2e}{dm^*} \frac{\int_{E_C}^\infty \tau_m (E-E_C)^{d/2} \frac{\partial f_0}{\partial E} dE}{\int_{E_C}^\infty f_0\,(E-E_C)^{\frac{d}{2}-1}\, dE}\right)}_{\mu_{dd}} \mathcal{F}_i \tag{8}$$

- **Mobility in d-dimension:**

$$\begin{aligned}
\mu_{dd} &= -\frac{2e}{dm^*}\frac{\int_{E_C}^{\infty}\tau_m(E-E_C)^{d/2}\frac{\partial f_0(E)}{\partial E}dE}{\int_{E_C}^{\infty}f_0(E)\times(E-E_C)^{\frac{d}{2}-1}\,dE} && (9)\\
&= -\frac{2e}{dm^*}\frac{\int_{E_C}^{\infty}\tau_m(E-E_C)^{d/2}\frac{\partial f_0(E)}{\partial E}\,dE}{\int_{E_C}^{\infty}f_0(E)\times(E-E_C)^{\frac{d}{2}-1}\,dE}\\
&= -\frac{2e}{dm^*}\frac{(k_BT)^{\frac{d}{2}}\int_0^{\infty}\tau_m\varepsilon^{d/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon}{(k_BT)^{\frac{d}{2}}\int_0^{\infty}f_0\times\varepsilon^{\frac{d}{2}-1}\,d\varepsilon}\\
&= -\frac{e\tau_0}{m^*}\frac{\int_0^{\infty}\varepsilon^{\frac{d}{2}+p}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon}{\frac{d}{2}\int_0^{\infty}f_0\times\varepsilon^{\frac{d}{2}-1}\,d\varepsilon}\;[\because\tau_m=\tau_0\times\varepsilon^p] && (10)\\
&= -\frac{e\tau_0}{m^*}\frac{-\Gamma(\frac{d}{2}+p+1)\,F_{\frac{d}{2}+p-1}(\eta_F)}{\frac{d}{2}\times\Gamma(\frac{d}{2})\times F_{\frac{d}{2}-1}(\eta_F)}\\
&= \frac{e\tau_0}{m^*}\left(\frac{\Gamma(\frac{d}{2}+p+1)}{\Gamma(\frac{d}{2}+1)}\right)\left(\frac{F_{\frac{d}{2}+p-1}(\eta_F)}{F_{\frac{d}{2}-1}(\eta_F)}\right) && (11)
\end{aligned}$$

Since, $f_0(\varepsilon)\to 0$ as $\varepsilon\to\infty$ and $\varepsilon^m\to 0$ as $\varepsilon\to 0$, using integration by parts $[\int f(x)g'(x)dx = f(x)g(x)-\int f'(x)g(x)dx]$

$$\begin{aligned}
&\int_0^{\infty}\varepsilon^{\frac{d}{2}+p}\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}\,d\varepsilon\\
&= [\varepsilon^{\frac{d}{2}+p}\,f_0(\varepsilon)]_0^{\infty}-\int_0^{\infty}\left(\frac{d}{2}+p\right)\times\varepsilon^{\frac{d}{2}+p-1}\,f_0(\varepsilon)\,d\varepsilon\\
&= -\left(\frac{d}{2}+p\right)\int_0^{\infty}\varepsilon^{\frac{d}{2}+p-1}\,f_0(\varepsilon)\,d\varepsilon\\
&= -\left(\frac{d}{2}+p\right)\,\Gamma(\frac{d}{2}+p)\,F_{\frac{d}{2}+p-1}(\eta_F)\\
&= -\Gamma(\frac{d}{2}+p+1)\,F_{\frac{d}{2}+p-1}(\eta_F) && (12)
\end{aligned}$$

- **In 3D:**

$$
\begin{aligned}
n_{3d} &= \left(\frac{2m^*\,k_BT}{\pi\hbar^2}\right)^{\frac{3}{2}}\,\frac{1}{4}\,F_{\frac{1}{2}}(\eta_F) \\
&= 2\left(\frac{m^*\,k_BT}{2\,\pi\hbar^2}\right)^{\frac{3}{2}}\,F_{\frac{1}{2}}(\eta_F) \\
&= N_C^{3d}\,F_{\frac{1}{2}}(\eta_F) \qquad (13)\\
N_C^{3d} &= 2\left(\frac{m^*\,k_BT}{2\,\pi\hbar^2}\right)^{\frac{3}{2}} \qquad (14)\\
\mu_{3d} &= \frac{e\tau_0}{m^*}\frac{\Gamma(\frac{5}{2}+p)}{\Gamma(\frac{5}{2})}\frac{F_{\frac{1}{2}+p}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)} \\
&= \frac{4e\tau_0}{3\sqrt{\pi}\,m^*}\frac{1}{F_{\frac{1}{2}}(\eta_F)}\Gamma(\frac{5}{2}+p)\times F_{\frac{1}{2}+p}(\eta_F) \qquad (15)
\end{aligned}
$$

3 Fermi-Dirac integral

- **Fermi-Dirac integral of order 0: (analytical solution)**

$$
F_0(\eta_F) = \frac{1}{\Gamma(1)}\int_0^\infty \frac{1}{1+e^{\varepsilon-\eta_F}}d\varepsilon = \ln[1+\exp(\eta_F)]
$$

- **Fermi-Dirac integral of order 1/2:**

$$
F_{\frac{1}{2}}(\eta_F) = \frac{1}{\Gamma(\frac{1}{2}+1)}\int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1+e^{\varepsilon-\eta_F}}d\varepsilon = \frac{2}{\sqrt{\pi}}\int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1+e^{\varepsilon-\eta_F}}d\varepsilon
$$

- **Inverse Fermi-Dirac integral of order 1/2:**

$$
\begin{aligned}
n_{3d} &= N_C^{3d}\,F_{\frac{1}{2}}(\eta_F) = N_C^{3d}\,\frac{2}{\sqrt{\pi}}\mathcal{F}_{\frac{1}{2}}(\eta_F) = N_{3d}\,\mathcal{F}_{\frac{1}{2}}(\eta_F) \\
N_{3d} &= \frac{2}{\sqrt{\pi}}\,N_C^{3d} = \frac{2}{\sqrt{\pi}}2\left(\frac{m^*\,k_BT}{2\,\pi\hbar^2}\right)^{\frac{3}{2}} = \frac{4}{\sqrt{\pi}}\left(\frac{m^*\,k_BT}{2\,\pi\hbar^2}\right)^{\frac{3}{2}} \\
\mathcal{F}_{\frac{1}{2}}(\eta_F) &= \int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1+e^{\varepsilon-\eta_F}}d\varepsilon
\end{aligned}
$$

1. **Using Joyce-Dixon approximation:**[8]

$$\begin{aligned}
\because n_{3d} &= N_C^{3d}\, F_{1/2}(\eta_{_F}) \\
\implies \eta_{_F} &= F_{1/2}^{-1}\left(\frac{n_{3d}}{N_C^{3d}}\right) \qquad (16)\\
\eta_{_F} &\simeq \ln\left(\frac{n_{3d}}{N_C^{3d}}\right) + \sum_{l=1}^{3} A_l \times \left(\frac{n_{3d}}{N_C^{3d}}\right)^l \qquad (17)\\
A_l &= 0.3535534, -4.95009\times 10^{-3}, 1.48386\times 10^{-4}
\end{aligned}$$

Note that the equation here is in terms of n_{3d} and N_C^{3d}.
To compare with the original derivation in Ref.[8] use this notation: $[\mathfrak{F} = 2\pi^{-1/2}\, F]_{publication} \equiv [F_{1/2} = 2\pi^{-1/2}\, \mathcal{F}_{1/2}]_{our}$.
Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
N_C^{3d} &= 2\left(\frac{m^*\, k_B T}{2\pi\hbar^2}\right)^{3/2} \\
&= 2\left(\frac{k_B\, m_0}{2\pi\hbar^2}\right)^{3/2} \times 10^{-6}\,\mathrm{cm}^{-3}\mathrm{K}^{-3/2} \times (m^{*'}\, T')^{3/2}\ \mathrm{K}^{3/2} \\
&= 4.829367 \times (m^{*'}\, T')^{3/2} \times 10^{15}\ \mathrm{cm}^{-3} \qquad (18)\\
\frac{n_{3d}}{N_C^{3d}} &= \frac{n'_{3d} \times 10^{18}\ \mathrm{cm}^{-3}}{4.829367 \times (m^{*'}\, T')^{3/2} \times 10^{15}\ \mathrm{cm}^{-3}} \\
&= 207.0665 \times \frac{n'_{3d}}{(m^{*'}\, T')^{3/2}} \qquad (19)
\end{aligned}$$

2. **Using minimax rational function approximation:**[9]

$$\begin{aligned}
\because n_{3d} &= N_{3d}\, \mathcal{F}_{1/2}(\eta_{_F}) \\
\implies \eta_{_F} &= \mathcal{F}_{1/2}^{-1}\left(\frac{n_{3d}}{N_{3d}}\right) \qquad (20)\\
\eta_{_F} &\simeq \text{Piecewise functions; details are in the Refeference.}
\end{aligned}$$

Note that the equation here is in terms of n_{3d} and N_{3d}.
Unit analysis (with 'prime' are unitless numbers; using units

from parameter table from Sec.1):

$$
\begin{aligned}
N_{3d} &= \frac{2}{\sqrt{\pi}} N_C^{3d} \\
&= \frac{2}{\sqrt{\pi}} \times 4.829367 \times (m^{*'}\, T')^{3/2} \times 10^{15}\ \mathrm{cm}^{-3} \qquad (21) \\
\frac{n_{3d}}{N_{3d}} &= \frac{n'_{3d} \times 10^{18}\ \mathrm{cm}^{-3}}{5.44936 \times (m^{*'}\, T')^{3/2} \times 10^{15}\ \mathrm{cm}^{-3}} \\
&= 183.50792 \times \frac{n'_{3d}}{(m^{*'}\, T')^{3/2}} \qquad (22)
\end{aligned}
$$

- **Fermi-Dirac integral of order 1:**

$$
\begin{aligned}
F_1(\eta_F) &= \frac{1}{\Gamma(2)} \int_0^\infty \frac{\varepsilon}{1+e^{\varepsilon-\eta_F}} d\varepsilon = \int_0^\infty \frac{\varepsilon}{1+e^{\varepsilon-\eta_F}} d\varepsilon \qquad (23) \\
F_1(\eta_F) &= -\mathrm{Li}_2(-e^{\eta_F}) \qquad (24)
\end{aligned}
$$

The integration can be calculated numerically using Eqn. 23 (implemented using `scipy.quad`) or using dilogarithm formalism of complete Fermi-Dirac integral of order-1, Eqn. 24 (implemented as `(-1) * scipy.spence(1+e^eta_f)`). Both are implemented in our package.

4 Scattering mechanisms

$$
\begin{aligned}
k_F &= \left(3\pi^2 n_{3D}\right)^{\frac{1}{3}} \qquad (25) \\
q_{TF}^{-1} &= \sqrt{\frac{2\varepsilon\epsilon_F}{3e^2 n_{3D}}} \qquad (26) \\
\varepsilon_F &= \frac{\hbar^2 k_F^2}{2\, m^*} \qquad (27)
\end{aligned}
$$

4.1 Structural related scattering

4.1.1 Alloy-disorder scattering

The implementation is based on Refs.[3, 4]. For alloy disorder mediated scattering $p = -\frac{1}{2}$. The momentum relaxation time:

$$\tau_{0,AD} = \frac{\hbar}{2\sqrt{\pi}} \frac{k_B\,T}{U_0^2\,\Omega_0\,x\,(1-x)\,N_C^{3d}} \tag{28}$$

$$\begin{aligned}
\mu_{3d,AD} &= \frac{4e\,\tau_{0,AD}}{3\,\sqrt{\pi}\,m^*} \frac{1}{F_{\frac{1}{2}}(\eta_F)} \Gamma(2) \times F_0(\eta_F) \text{ [From Eqn.15]} \\
&= \frac{2e\hbar}{3\pi} \frac{k_B\,T}{m^*\,U_0^2\,\Omega_0\,x\,(1-x)} \frac{1}{N_C^{3d}\,F_{\frac{1}{2}}(\eta_F)} F_0(\eta_F) \\
&= \frac{2e\hbar}{3\pi} \frac{k_B\,T}{m^*\,U_0^2\,\Omega_0\,x\,(1-x)\,n_{3d}} F_0(\eta_F) \text{ [From Eqn.13]} \qquad (29)
\end{aligned}$$

$$(30)$$

Re-writing as function position, z:

$$\mu_{3d,AD}(z) = \frac{2e\hbar}{3\pi} \frac{k_B\,T}{m^*(z)\,U_0(z)^2\,\Omega_0(z)\,x(z)\,(1-x(z))\,n_{3d}(z)} F_0(\eta_F(z)) \tag{31}$$

$$F_0(\eta_F(z)) = \ln[1 + \exp(\eta_F(z))] \tag{32}$$

$$\eta_F(z) = \ln\left(\frac{n_{3d}(z)}{N_C^{3d}(z)}\right) + \sum_{l=1}^{3} A_l \times \left(\frac{n_{3d}(z)}{N_C^{3d}(z)}\right)^l \tag{33}$$

$$A_l = 0.3535534, -4.95009 \times 10^{-3}, 1.48386 \times 10^{-4}$$

$$N_C^{3d}(z) = 2\left(\frac{k_B T}{2\pi\hbar^2} m^*(z)\right)^{3/2} \tag{34}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$
\begin{aligned}
N_C^{3d}(z) &= 2\left(\frac{k_B T}{2\pi\hbar^2} m^*(z)\right)^{3/2} \\
&= 2\left(\frac{k_B\, m_0}{2\pi\hbar^2}\right)^{3/2} \times 10^{-6}\,\mathrm{cm}^{-3}\mathrm{K}^{-3/2} \times [m^{*'}(z)\, T']^{3/2}\ \mathrm{K}^{3/2} \\
&= 4.829367 \times [m^{*'}(z)\, T']^{3/2} \times 10^{15}\ \mathrm{cm}^{-3} \qquad (35) \\
\frac{n_{_{3d}}(z)}{N_C^{3d}(z)} &= \frac{n'_{_{3d}}(z) \times 10^{18}\ \mathrm{cm}^{-3}}{4.829367 \times [m^{*'}(z)\, T']^{3/2} \times 10^{15}\ \mathrm{cm}^{-3}} \\
&= 207.0665 \times \frac{n'_{_{3d}}(z)}{[m^{*'}(z)\, T']^{3/2}} \qquad (36) \\
\mu_{_{3d,AD}}(z) &= \frac{2e\hbar}{3\pi} \frac{k_B\, T}{m^*(z)\, U_0(z)^2\, \Omega_0(z)\, x(z)\, (1 - x(z))\, n_{_{3d}}(z)} F_0(\eta_{_F}(z)) \\
&= \frac{2e\hbar\, k_B\, 10^4}{3\pi m_0\, e^2\, (10^{-8})^3\, 10^{18}} \frac{T}{m^*(z)\, U_0(z)^2\, \Omega_0(z)\, x(z)\, (1 - x(z))\, n_{_{3d}}(z)} \\
&\qquad \times F_0(\eta_{_F}(z))\ \mathrm{cm}^2\, \mathrm{V}^{-1}\, \mathrm{s}^{-1}\, \mathrm{K}^{-1} \\
&= 21.16991 \times \frac{T'}{m^{*'}(z)\, U_0'(z)^2\, \Omega_0'(z)\, x(z)\, (1 - x(z))\, n'_{_{3d}}(z)} \\
&\qquad \times F_0(\eta_{_F}(z))\ \mathrm{cm}^2\, \mathrm{V}^{-1}\, \mathrm{s}^{-1}\, \mathrm{K}^{-1} \qquad (37)
\end{aligned}
$$

4.1.2 Dislocation momentum relaxation rate

4.1.2.1 Scattering from line charge of threading dislocation core

$$
\frac{1}{\tau_{_{DIS}}} = \frac{m^* e^4 N_{_{DIS}} f_{_{DIS}}^2}{4\pi k_{_F}^4 c_0^2 \varepsilon_s^2 \hbar^3} \times \int_0^1 \frac{1}{\left(u + \frac{q_{_{TF}}}{2k_{_F}}\right)^2 \sqrt{1 - u^2}} du \qquad (38)
$$

4.1.2.2 Scattering from strain field of threading dislocation

This scattering mechanism is implemented based on Refs. [10, 3, 11]. This form of scattering arises due to the presence of dislocations, regardless of the presence or absence of charges at the core.

2.1.2.2.A Due to threading edge dislocation strain field

$$\frac{1}{\tau_{DIS}^{str}} = \frac{m^* N_{DIS} b_e^2 E_D^2}{2\pi k_F^2 \hbar^3} \left(\frac{1-2\gamma}{1-\gamma}\right)^2 \times \int_0^1 \frac{u^2}{\left(u + \frac{q_{TF}}{2k_F}\right)^2 \sqrt{1-u^2}} du \quad (39)$$

$$b_e = \frac{1}{3} < 11\overline{20} >= a_0 \quad (40)$$

$$\gamma = \frac{C_{12}}{C_{11}+C_{12}} \quad [\text{“Isotropic Poisson's ratio”}] \quad (41)$$

NOTE-1: For WZ structure the Poisson ratio is anisotropic[12]. However, the strain field due to edge dislocation uses 'isotropic/average Poisson ratio'. We use the definition $\gamma = C_{12}/(C_{11}+C_{12})$ following reference Shi et al., APL 74, 573 (1999)[11]. This was also used in the thesis.[3]
NOTE-2: In Ref.[12] there is a printing mistake. The s_{11} expression should be $2s_{11} = sc_{33} + \frac{1}{c_{11}-c_{12}}$.

2.1.2.2.B Due to threading screw dislocation strain field
Screw dislocation strain scattering has not yet been implemented.

4.2 Phonon related scattering

4.2.1 Deformation potential scattering

The implementation is based on Refs.[3]. For deformation potential acoustic phonon mediated scattering $p = -\frac{1}{2}$. The momentum relaxation time:

$$\tau_{0,DP} = \frac{\hbar}{2\sqrt{\pi}} \frac{\rho\, v_{LA}^2}{E_D^2\, N_C^{3d}} \quad (42)$$

$$\mu_{3d,DP} = \frac{4e\,\tau_{0,DP}}{3\sqrt{\pi}\, m^*} \frac{1}{F_{\frac{1}{2}}(\eta_F)} \Gamma(2) \times F_0(\eta_F) \text{ [From Eqn.15]}$$

$$= \frac{4e\hbar}{6\pi\, m^*} \frac{\rho\, v_{LA}^2}{E_D^2} \frac{1}{N_C^{3d}\, F_{\frac{1}{2}}(\eta_F)} F_0(\eta_F)$$

$$= \frac{2e\hbar}{3\pi\, m^*} \frac{\rho\, v_{LA}^2}{E_D^2\, n_{3d}} F_0(\eta_F) \text{ [From Eqn.13]} \quad (43)$$

Re-writing as function position, z:

$$\mu_{3d,DP}(z) = \frac{2e\hbar}{3\pi}\frac{\rho(z)\,v_{LA}(z)^2}{m^*(z)\,E_D(z)^2\,n_{3d}(z)}F_0(\eta_F(z)) \tag{44}$$

$$F_0(\eta_F(z)) = \ln[1+\exp(\eta_F(z))] \tag{45}$$

$$\eta_F(z) = \ln\left(\frac{n_{3d}(z)}{N_C^{3d}(z)}\right) + \sum_{l=1}^{3} A_l \times \left(\frac{n_{3d}(z)}{N_C^{3d}(z)}\right)^l \tag{46}$$

$$A_l = 0.3535534, -4.95009\times10^{-3}, 1.48386\times10^{-4}$$

$$N_C^{3d}(z) = 2\left(\frac{k_BT}{2\pi\hbar^2}m^*(z)\right)^{3/2} \tag{47}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned} N_C^{3d}(z) &= 2\left(\frac{k_BT}{2\pi\hbar^2}m^*(z)\right)^{3/2} \\ &= 2\left(\frac{k_B\,m_0}{2\pi\hbar^2}\right)^{3/2}\times10^{-6}\,\mathrm{cm}^{-3}\mathrm{K}^{-3/2}\times[m^{*'}(z)\,T']^{3/2}\,\mathrm{K}^{3/2} \\ &= 4.829367\times[m^{*'}(z)\,T']^{3/2}\times10^{15}\,\mathrm{cm}^{-3} \end{aligned} \tag{48}$$

$$\begin{aligned} \frac{n_{3d}(z)}{N_C^{3d}(z)} &= \frac{n'_{3d}(z)\times10^{18}\,\mathrm{cm}^{-3}}{4.829367\times[m^{*'}(z)\,T']^{3/2}\times10^{15}\,\mathrm{cm}^{-3}} \\ &= 207.0665\times\frac{n'_{3d}(z)}{[m^{*'}(z)\,T']^{3/2}} \end{aligned} \tag{49}$$

$$\begin{aligned} \mu_{3d,DP}(z) &= \frac{2e\hbar}{3\pi}\frac{\rho(z)\,v_{LA}(z)^2}{m^*(z)\,E_D(z)^2\,n_{3d}(z)}F_0(\eta_F(z)) \\ &= \frac{2e\hbar\,10^{-2}}{3\pi m_0\,e^2\,10^{18}}\frac{\rho'(z)\,v'_{LA}(z)^2}{m^{*'}(z)\,E'_D(z)^2\,n'_{3d}(z)}F_0(\eta_F(z))\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1} \\ &= 1.5333\times10^{-6}\frac{\rho'(z)\,v'_{LA}(z)^2}{m^{*'}(z)\,E'_D(z)^2\,n'_{3d}(z)}F_0(\eta_F(z))\,\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1} \end{aligned} \tag{50}$$

4.2.2 Piezoelectric phonon scattering

The implementation is based on Refs.[3, 2]. Note following [2] we use K^2 term in our below derivation which is related to the terms in Ref.[3] as:

$$\rho v_s^2 \equiv C^*; K^2 \equiv \langle \frac{h_{pz}^2}{C^* \epsilon_s} \rangle$$

For piezoelectric phonon scattering $p = +\frac{1}{2}$. The momentum relaxation time:

$$\begin{aligned}
\tau_{_{0,PE}} &= \frac{2m^* \epsilon_s \, k_B T}{\sqrt{\pi} \, \hbar \, e^2 \, K^2} \frac{1}{N_C^{3d}} && (51) \\
\mu_{_{3d,PE}} &= \frac{4e \, \tau_{_{0,PE}}}{3 \sqrt{\pi} \, m^*} \frac{1}{F_{\frac{1}{2}}(\eta_{_F})} \Gamma(3) \times F_1(\eta_{_F}) \text{ [From Eqn.15]} \\
&= \frac{16 \epsilon_s \, k_B T}{3\pi\hbar e K^2} \frac{1}{N_C^{3d} \, F_{\frac{1}{2}}(\eta_{_F})} F_1(\eta_{_F}) \\
&= \frac{16 \, k_B T}{3\pi\hbar e} \frac{\epsilon_s}{K^2 \, n_{_{3d}}} F_1(\eta_{_F}) \text{ [From Eqn.13]} && (52)
\end{aligned}$$

Re-writing as function position, z:

$$\begin{aligned}
\mu_{_{3d,PE}}(z) &= \frac{16 \, k_B T}{3\pi\hbar e} \frac{\epsilon_s(z)}{K^2(z) \, n_{_{3d}}(z)} F_1(\eta_{_F}(z)) && (53) \\
F_1(\eta_{_F}(z)) &= \int_0^\infty \frac{\varepsilon}{1 + e^{\varepsilon - \eta_{_F}(z)}} d\varepsilon && (54) \\
\eta_{_F}(z) &= \ln\left(\frac{n_{_{3d}}(z)}{N_C^{3d}(z)}\right) + \sum_{l=1}^{3} A_l \times \left(\frac{n_{_{3d}}(z)}{N_C^{3d}(z)}\right)^l && (55) \\
A_l &= 0.3535534, -4.95009 \times 10^{-3}, 1.48386 \times 10^{-4} \\
N_C^{3d}(z) &= 2\left(\frac{k_B T}{2\pi\hbar^2} m^*(z)\right)^{3/2} && (56)
\end{aligned}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$
\begin{aligned}
N_C^{3d}(z) &= 2\left(\frac{k_B T}{2\pi\hbar^2} m^*(z)\right)^{3/2} \\
&= 2\left(\frac{k_B\, m_0}{2\pi\hbar^2}\right)^{3/2} \times 10^{-6}\,\mathrm{cm}^{-3}\mathrm{K}^{-3/2} \times [m^{*'}(z)\, T']^{3/2}\ \mathrm{K}^{3/2} \\
&= 4.829367 \times [m^{*'}(z)\, T']^{3/2} \times 10^{15}\ \mathrm{cm}^{-3} \qquad (57) \\
\frac{n_{3d}(z)}{N_C^{3d}(z)} &= \frac{n'_{3d}(z) \times 10^{18}\ \mathrm{cm}^{-3}}{4.829367 \times [m^{*'}(z)\, T']^{3/2} \times 10^{15}\ \mathrm{cm}^{-3}} \\
&= 207.0665 \times \frac{n'_{3d}(z)}{[m^{*'}(z)\, T']^{3/2}} \qquad (58) \\
\mu_{3d,PE}(z) &= \frac{16\, k_B}{3\pi\hbar e} \frac{\epsilon_s(z)\, T}{K^2(z)\, n_{3d}(z)} F_1(\eta_F(z)) \\
&= \frac{16\, k_B \epsilon_0}{3\pi\hbar e} \frac{\epsilon'_s(z)\, T}{K^{2'}(z)\, n'_{3d}(z) \times 10^{18}} F_1(\eta_F(z))\ \mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}\,\mathrm{K}^{-1} \\
&= 0.12283 \times \frac{\epsilon'_s(z)\, T'}{K^{2'}(z)\, n'_{3d}(z)} F_1(\eta_F(z))\ \mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1} \qquad (59)
\end{aligned}
$$

4.2.3 Polar optical phonon scattering

The implementation is based on Refs.[3, 4]. For polar optical phonon mediated scattering, the mobility is:

$$
\begin{aligned}
\mu_{0,POP} &= \frac{4\pi\epsilon^*\hbar}{e\, m^* q_0} \frac{1}{N_{BE}} \\
&= \frac{2^{\frac{3}{2}}\pi\epsilon^*\hbar^2}{e\, m^{*\frac{3}{2}} E_{pop}^{\frac{1}{2}}} \times \left(e^{\frac{E_{pop}}{k_B T}} - 1\right) \qquad (60) \\
\frac{1}{\epsilon^*} &= \frac{1}{\epsilon_h} - \frac{1}{\epsilon_s} \qquad (61) \\
\hbar\omega_{pop} &= \frac{\hbar^2 q_0^2}{2m^*} \implies q_0 = \sqrt{\frac{2m^*\, E_{POP}}{\hbar^2}} \\
N_{BE} &= \frac{1}{e^{\frac{\hbar\omega_{pop}}{k_B T}} - 1}\ \text{[Bose-Einstein distribution]} \qquad (62)
\end{aligned}
$$

Re-writing as function position, z:

$$\mu_{3d,POP}(z) = \frac{2^{\frac{3}{2}}\pi\hbar^2}{e}\frac{\epsilon^*(z)}{m^*(z)^{\frac{3}{2}}E_{pop}(z)^{\frac{1}{2}}} \times \left(e^{\frac{E_{pop}(z)}{k_B T}} - 1\right) \tag{63}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\mu_{3d,POP}(z) &= \frac{2^{\frac{3}{2}}\pi\hbar^2}{e}\frac{\epsilon^*(z)}{m^*(z)^{\frac{3}{2}}E_{pop}(z)^{\frac{1}{2}}} \times \left(e^{\frac{E_{pop}(z)}{k_B T}} - 1\right) \\
&= \frac{2^{\frac{3}{2}}\pi\hbar^2\epsilon_0 \times 10^4}{e \times m_0^{3/2} \times e^{1/2}}\frac{\epsilon^{*'}(z)}{m^{*'}(z)^{\frac{3}{2}}E'_{pop}(z)^{\frac{1}{2}}} \times \left(e^{\frac{e\times E'_{pop}(z)}{k_B T'}} - 1\right) \\
&\qquad \mathrm{cm^2\,V^{-1}\,s^{-1}} \\
&= 0.15693 \times \frac{\epsilon^{*'}(z)}{m^{*'}(z)^{\frac{3}{2}}E'_{pop}(z)^{\frac{1}{2}}} \times \left(e^{11604.518\times\frac{E'_{pop}(z)}{T'}} - 1\right) \\
&\qquad \mathrm{cm^2\,V^{-1}\,s^{-1}}
\end{aligned} \tag{64}$$

References

[1] Badal Mondal. Semiconductortransport/mobilitypy: version-1.0.5, 2026.

[2] Chihiro Hamaguchi. *Electron–Phonon Interaction and Electron Transport*, pages 273–374. Springer International Publishing, Cham, 2023.

[3] Debdeep Jena. *Polarization induced electron populations in III-V nitride semiconductors transport, growth, and device applications*. PhD dissertation, UNIVERSITY of CALIFORNIA Santa Barbara, 2003.

[4] Siddharth Rajan, Steven P. DenBaars, Umesh K. Mishra, Huili (Grace) Xing, and Debdeep Jena. Electron mobility in graded algan alloys. *Applied Physics Letters*, 88:1–3, 1 2006.

[5] Debdeep Jena, Arthur C. Gossard, and Umesh K. Mishra. Dislocation scattering in a two-dimensional electron gas. *Applied Physics Letters*, 76:1707–1709, 3 2000.

[6] Debdeep Jena, Sten Heikman, James S. Speck, Arthur Gossard, Umesh K. Mishra, Angela Link, and Oliver Ambacher. Magnetotransport properties of a polarization-doped three-dimensional electron slab in graded algan. *Physical Review B*, 67:153306, 4 2003.

[7] Debdeep Jena and Umesh K. Mishra. Quantum and classical scattering times due to charged dislocations in an impure electron gas. *Physical Review B*, 66:241307, 12 2002.

[8] W. B. Joyce and R. W. Dixon. Analytic approximations for the fermi energy of an ideal fermi gas. *Applied Physics Letters*, 31(5):354–356, 09 1977.

[9] Toshio Fukushima. Precise and fast computation of inverse Fermi-Dirac integral of order 1/2 by minimax rational function approximation. *Applied Mathematics and Computation*, 259:698–707, 2015.

[10] D. Jena, I. Smorchkova, A.C. Gossard, and U.K. Mishra. Electron transport in III-V Nitride two-dimensional electron gases. *physica status solidi (b)*, 228:617–619, 11 2001.

[11] Changchun Shi, Peter M. Asbeck, and Edward T. Yu. Piezoelectric polarization associated with dislocations in wurtzite GaN. *Applied Physics Letters*, 74(4):573–575, 01 1999.

[12] R. Thokala and J. Chaudhuri. Calculated elastic constants of wide band gap semiconductor thin films with a hexagonal crystal structure for stress problems. *Thin Solid Films*, 266(2):189–191, 1995.